

1. Chapter 1 - 9, 11, 13, 14, 16, 18, 19, 20 and 26
2. chapter 2 - 5, 6, 7, 9, 11, 12, 13, 17, 22, 28 and 101
3. Chapter 3 - 3.2, 3.5, 3.9, 3.11
4. Chapter 3 - 3.12, 3.17, 3.19, 3.36, 3.37, 3.57, 3.60, 3.67, 3.72, 3.74
5. Chapter 4 - 4.5, 4.10, 4.15, 4.17, 4.18, 4.21, 4.24 (which is called Dini's theorem), 4.35, 4.36 and 4.43
6. Chapter 4 - 4.27, 4.30 and 4.41; and problems 1, 2, 12 and 13 from the "More Prelim Problems" section of Chapter 4; and chapter 5 - 5.1, 5.2 and 5.8
7. Chapter 5 - 5.15, 5.17, 5.18, 5.20, 5.22, 5.30; extra exercise: prove that the conclusion of the Inverse Function theorem does not necessarily hold if f is not of class C^1 , that is, find a differentiable function f , whose derivative is invertible at zero, but there is no local inverse for f in any neighborhood of zero.
8. HW 8 pdf attached